Palmer Penguins Part 5: Random Forest versus Linear Models

Settling the interpretability-performance tradeoff in ecological modelling

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Figure 1: Penguins on rocky Antarctic terrain

Palmer Penguins series, Part 5 (finale). Photo used under open licence.

Palmer Penguins Data Analysis Series

This is **Part 5** (finale) of a 5-part series exploring penguin morphometrics:

1. Part 1: EDA and Simple Regression
2. Part 2: Multiple Regression and Species Effects
3. Part 3: Cross-Validation and Model Comparison
4. Part 4: Model Diagnostics and Interpretation
5. **Part 5: Random Forest versus Linear Models** (this post)

1 Introduction

Whether a random forest meaningfully outperforms a linear model remains unclear until a careful head-to- head comparison is conducted. Throughout this series, the species-aware linear model has consistently delivered an R-squared of approximately 0.86, survived cross-validation with minimal degradation, and passed every diagnostic check devised. The natural question for the finale was whether a more flexible algorithm could do substantially better.

The answer is somewhat unexpected. The random forest improved R-squared by roughly two percentage points, from about 0.86 to about 0.88, despite having access to additional predictors and making no assumptions about linearity. That narrow gap prompts careful reflection on when additional model complexity is worth the cost in interpretability, reproducibility, and communication.

This post presents the full comparison, examines feature importance from both perspectives, and concludes the series with practical guidance for choosing between linear and ensemble methods in ecological research.

1.1 Motivations

- To settle the question of whether machine learning methods meaningfully outperform linear regression on this ecological dataset.
- To establish a principled framework for evaluating the interpretability-performance tradeoff, not just an intuitive judgement.
- To determine whether the random forest identifies the same important predictors as the linear model or reveals hidden structure.
- To provide a concrete example of a case where the simpler model is the better choice, as a counterpoint to the assumption that more complex methods are always superior.
- To summarise the entire five-part journey into a set of practical recommendations.
- To produce a reusable template for model comparison applicable to future datasets.

1.2 Objectives

1. Train a random forest with all available predictors and compare its cross-validated RMSE and R-squared against the species-aware linear model under identical evaluation conditions.
2. Extract and compare feature importance rankings from both models, and assess whether they align with biological expectations.
3. Develop a decision framework that maps project requirements (interpretability, sample size, regulatory context) to an appropriate model class.
4. Summarise the key findings and methodological lessons from all five parts of the series.

Errors and better approaches are welcome; see the Feedback section at the end.

Visual interlude before the technical content.

2 Prerequisites and Setup

This analysis loads the same core packages used throughout the series, with `randomForest` and `caret` providing the ensemble learning and cross-validation infrastructure.

```
library(palmerpenguins)
library(tidyverse)
library(broom)
library(caret)
library(randomForest)
library(knitr)
library(patchwork)

theme_set(theme_minimal(base_size = 12))
penguin_colors <- c(
  "Adelie" = "#FF6B6B",
  "Chinstrap" = "#4ECDC4",
  "Gentoo" = "#45B7D1"
)

data(penguins)
penguins_clean <- penguins |> drop_na()

set.seed(42)
```

We use the same random seed and cross-validation configuration as Part 3 to ensure that the results are directly comparable.

3 What is the Interpretability-Performance Tradeoff?

The interpretability-performance tradeoff is the observation that, in statistical modelling, gains in predictive accuracy often come at the expense of the analyst's ability to explain how the model arrives at its predictions. A linear regression produces a simple equation: body mass equals a weighted sum of morphometric measurements plus species offsets. Every coefficient has a direct biological interpretation, and confidence intervals quantify uncertainty in each estimate.

A random forest, by contrast, aggregates predictions from hundreds of decision trees, each built on a bootstrap sample with a random subset of predictors. The resulting predictions are often more accurate, but there is no single equation to inspect. Feature importance scores indicate which variables matter most, but they do not tell you the direction or magnitude of each effect in the way that regression coefficients do.

The tradeoff is not purely technical. In regulatory settings, grant applications, and peer-reviewed publications, reviewers and stakeholders often require models whose logic can be stated explicitly. A two-percentage-point improvement in R-squared may not justify the loss of that transparency.

4 Getting Started

4.1 Cross-Validation Configuration

We establish a single cross-validation scheme that both models will share, eliminating any advantage from different fold assignments.

```
train_control <- trainControl(
  method = "cv",
  number = 10,
  savePredictions = "final",
  verboseIter = FALSE
)
```

4.2 Training the Linear Champion

The species-aware linear model uses the same formula that has been our workhorse since Part 2.

```
cv_linear <- train(
  body_mass_g ~ bill_length_mm + bill_depth_mm +
    flipper_length_mm + species,
  data = penguins_clean,
  method = "lm",
  trControl = train_control
)
```

4.3 Training the Random Forest Challenger

The random forest receives all available predictors, including sex and island, which were not used in the linear model. This gives the ensemble every opportunity to outperform.

```
set.seed(123)

cv_rf <- train(
```